Meme Caption Generation Using LSTM and Transformers and Their Comparison

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Abstract

We study image-to-caption generation for meme templates on memes900k (300 templates, 900,000 captions), following DeepHumor. We run a controlled 2×2 ablation over two frozen visual encoders (ResNet-50 and CLIP ViT-B/32) and two decoders (LSTM and Transformer), reporting test-set BLEU and perplexity under a shared greedy decoder. At one epoch the decoder dominates: swapping the LSTM for a Transformer adds +5.65 BLEU with CLIP and +12.37 with ResNet-50, whereas the encoder choice only matters when the decoder is weak. After three epochs the Transformer runs cluster tightly (14.38–14.66 BLEU) and both encoder and embedding size wash out. Perplexity roughly halves with additional training while BLEU stays flat, exposing the exposure-bias gap between teacher-forced fitting and free-running generation. Our contribution is a clean, reproducible measurement methodology rather than a new state of the art.

1. Introduction

Image captioning—describing an image in natural language—is a core vision-language task. Meme templates are a constrained but semantically rich instance: a single image admits many valid captions, and the "humour" rarely follows the surface form of a reference caption. Borovik et al. (2020) introduced DeepHumor for memes900k (300 templates, 900,000 captions), but its public code reports only teacher-forced perplexity, never evaluates the test split, and its "beam search" is in fact unseeded multinomial sampling. This makes quantitative comparison difficult.

Goal. We provide a rigorous, controlled comparison that isolates the visual encoder (ResNet-50 (He et al., 2016) vs. CLIP (Radford et al., 2021)) and the decoder (LSTM (Hochreiter & Schmidhuber, 1997) vs. Transformer (Vaswani et al., 2017)), using the same decoding algorithm and metrics, and evaluating generated captions on the official test split.

Contributions.

- A 2×2 encoder×decoder ablation with frozen encoders and a single shared greedy decoder (Section 3).
- Test-set BLEU (Papineni et al., 2002; Post, 2018) alongside perplexity, at two budgets: 1 epoch (6,250 steps) and 3 epochs (18,750 steps).
- A rendered caption gallery and a one-command inference demo for qualitative inspection.
- The finding that the decoder architecture dominates, while behind a Transformer decoder the encoder and embedding size wash out; perplexity and BLEU decouple.

2. Background and Related Work

Early neural captioners pair CNN features with RNN decoders (Show and Tell, Vinyals et al., 2015); attention over spatial features improved grounding (Show, Attend and Tell, Xu et al., 2015). The Transformer (Vaswani et al., 2017) replaced recurrence with self- and cross-attention and became a standard decoder. Large-scale vision-language pretraining such as CLIP (Radford et al., 2021) provides semantically rich representations that transfer well to captioning.

Borovik et al. (2020) trained frozen ResNet-50 + LSTM (and a Transformer variant) on memes900k, using teacher-forced perplexity as the primary metric and an unseeded sampling decoder. We adopt the dataset and the frozen-feature paradigm but require

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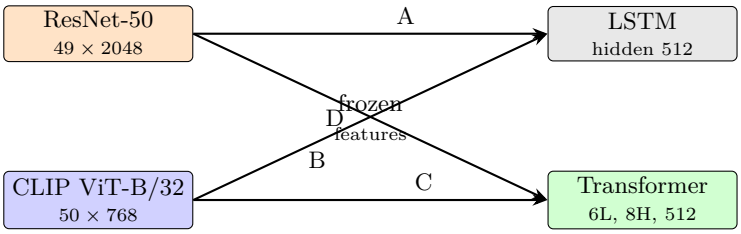


Figure 1. The 2×2 encoder×decoder design. All encoders are frozen and their features precomputed. Runs: A = ResNet-50+LSTM, B = CLIP+LSTM, C = CLIP+Transformer, D = ResNet-50+Transformer; E = C with embedding 768.

Table 1. Official caption-level split of memes900k.

Split	Rows	Per template
Train	750,000	2,500
Validation	75,000	250
Test	75,000	250

test-set BLEU with a shared greedy decoder and explicit ablations. BLEU (Papineni et al., 2002) measures n-gram overlap with references; Post (2018) recommend clear reporting (we use sacrebleu). Perplexity under teacher forcing is informative within a tokenizer family but not comparable across vocabularies.

3. Method

3.1. Corpus

memes900k provides 300 meme-template images and 900,000 tab-separated caption rows. We use the official caption-level split (Table 1): 750,000 train, 75,000 validation, 75,000 test. Every template appears in all splits; scores therefore reflect new captions for seen templates, not unseen-template generalization.

3.2. Architectures

Encoders (frozen). ResNet-50 (He et al., 2016) yields a 49×2048 spatial sequence; CLIP ViT-B/32 (Radford et al., 2021) yields a 50×768 patch-token sequence. Features are precomputed once per template, so training never decodes JPEGs.

Decoders. The LSTM (Hochreiter & Schmidhuber, 1997) (hidden 512, 2 layers, embedding 256) receives mean-pooled image features as its initial state. The Transformer (Vaswani et al., 2017) (6 layers, 8 heads, hidden 512, FFN 2048, embedding 512) attends to the visual tokens as encoder memory and ties its input and output embeddings.

The 2×2 design crosses each encoder with each decoder (Figure 1); runs are labelled A–D, with E the CLIP+Transformer embedding-768 variant.

3.3. Training and Budgets

All models share the optimizer and schedule: AdamW, learning rate 3×10^{-4}, warmup 500 steps, weight decay 10^{-4}, gradient accumulation 4, batch size 32, seed 42, and a 200,000-row training subsample. One epoch is therefore $200{,}000/32 = 6{,}250$ steps. Tier 1 uses 1 epoch (6,250 steps); Tier 2 uses 3 epochs (18,750 steps), matching the long DeepHumor run. Hyperparameters were screened at 6,000 steps by validation loss (one knob changed at a time), and the winning configuration was confirmed at full budget; Section 4 reports the screen–confirm protocol. Figure 2 shows the validation and training loss curves.

3.4. Metrics

Perplexity: $\text{ppl} = \exp(\overline{\text{NLL}})$ under teacher forcing over all 75,000 test rows; comparable only within this project's GPT-2 BPE vocabulary. BLEU: sacrebleu corpus BLEU on a fixed, seeded 5,000-row test sample with greedy decoding (Papineni et al., 2002; Post, 2018)—the cross-tokenizer comparative number.

4. Results

4.1. Tier 1 — 1 epoch (6,250 steps)

Table 2 reports Tier 1. The decoder dominates: LSTM→Transformer gives +5.65 BLEU (B→C) and +12.37 (A→D). The encoder matters only when the decoder is weak (+6.79 for A→B) and is flat behind a Transformer (C→D = −0.07). A larger embedding (E) improves perplexity but not BLEU.

4.2. Tier 2 — 3 epochs (18,750 steps)

Table 3 reports Tier 2. The Transformer runs form a tight cluster (14.38–14.66), and both LSTM runs remain behind (12.16–13.36) even though they improve substantially over Tier 1. Perplexity roughly halves relative to Tier 1 while BLEU is essentially unchanged

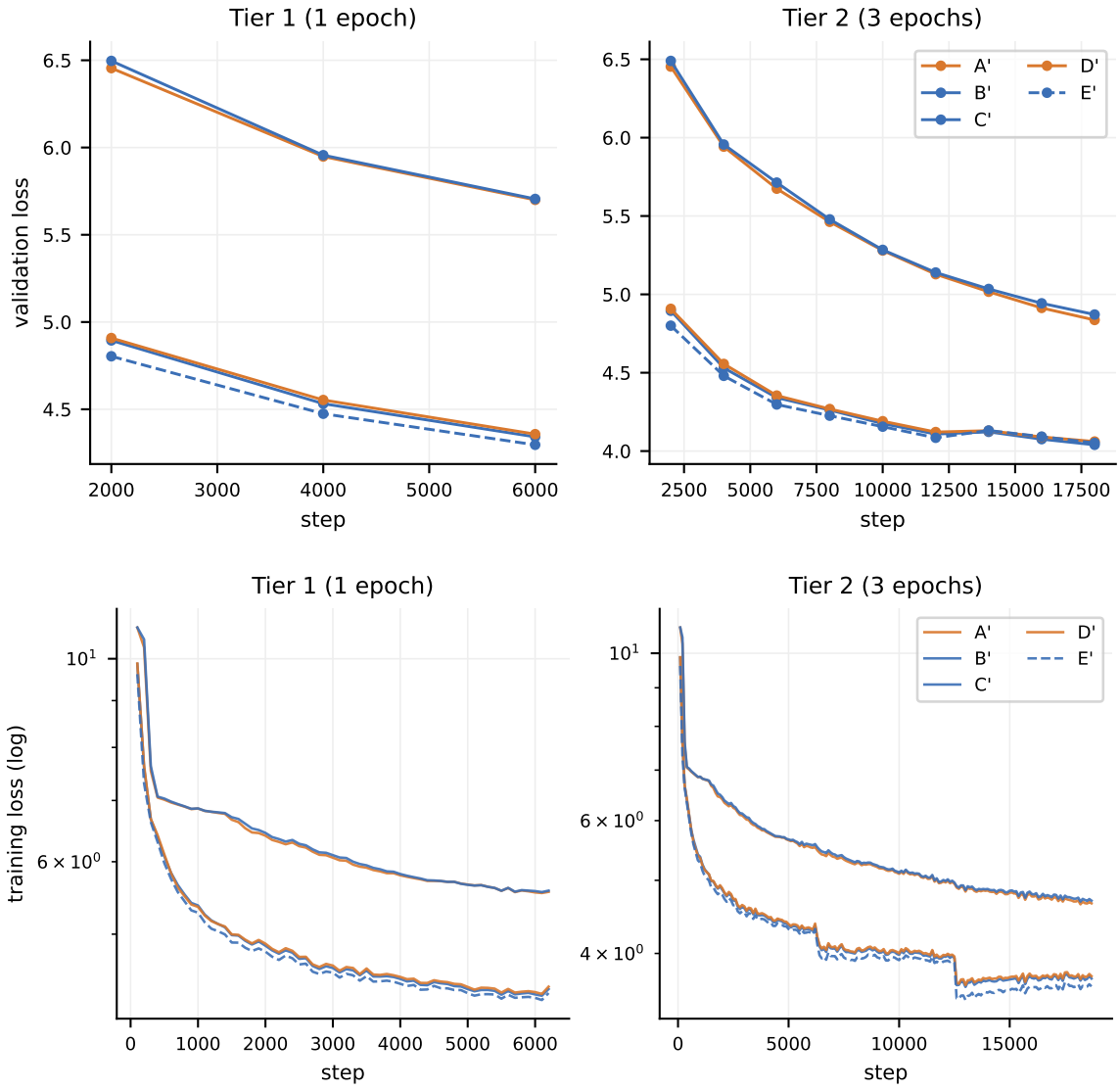


Figure 2. Training dynamics. Top: validation loss; bottom: training loss (log scale). Tier 1 (left) and Tier 2 (right). The Transformer runs converge faster and lower.

Table 2. Tier 1 results (1 epoch, official test split).

Run	Encoder	Decoder	BLEU	Ppl
A	ResNet-50	LSTM	2.13	255.5
B	CLIP	LSTM	8.92	259.7
C	CLIP	Transformer	14.57	71.5
D	ResNet-50	Transformer	14.50	72.7
E	CLIP	Transformer (768)	14.36	68.4

Table 3. Tier 2 results (3 epochs, official test split).

Run	Encoder	Decoder	BLEU	Ppl
A′	ResNet-50	LSTM	13.36	115.9
B′	CLIP	LSTM	12.16	120.7
C′	CLIP	Transformer	14.38	54.7
D′	ResNet-50	Transformer	14.63	55.5
E′	CLIP	Transformer (768)	14.66	55.2

(Figure 3).

4.3. Hyperparameter screening

Figure 4 ranks the six screening configurations by validation loss at 6,000 steps. Increasing the embedding to 768 is the only change that beats the base configu-

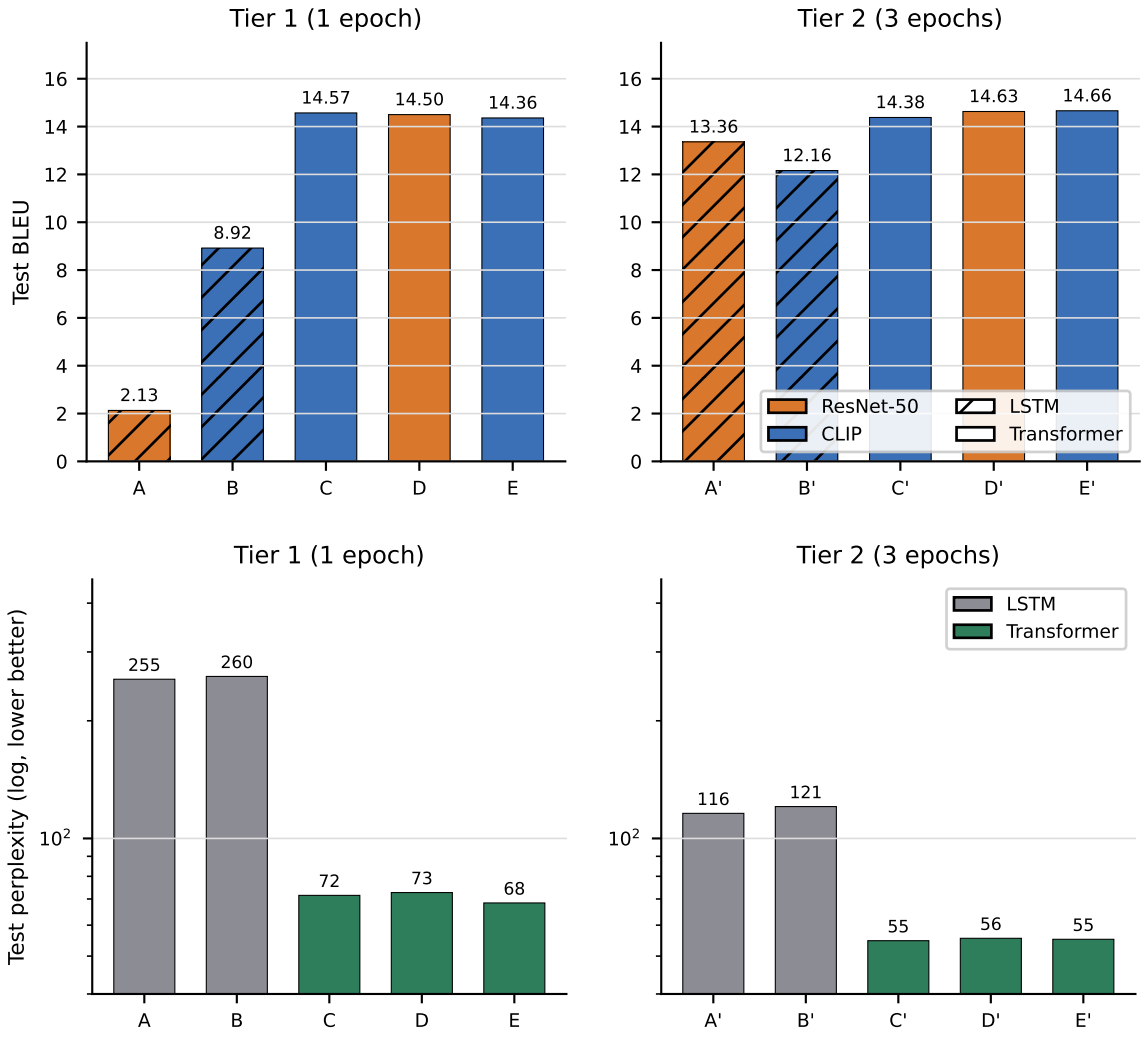


Figure 3. Test metrics by run for each tier. Top: BLEU (bars coloured by encoder, hatched by decoder). Bottom: perplexity (log scale, lower is better). The Transformer runs dominate in both tiers.

ration; the remaining knobs are within noise of it.

4.4. Perplexity vs. BLEU

Figure 5 plots BLEU against perplexity for all runs. Perplexity varies over an order of magnitude while BLEU saturates around 14.5, and the 1-epoch and 3-epoch Transformer runs overlap despite very different losses. The validation-loss ranking does not match the BLEU ranking—the exposure-bias gap.

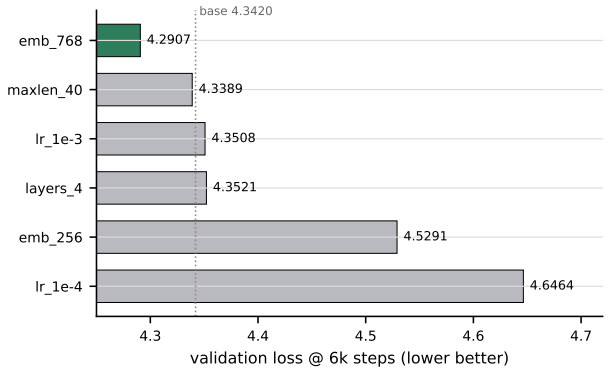


Figure 4. One-knob-at-a-time screening at 6,000 steps. Validation loss (lower is better); the embed-768 variant (highlighted) is the winner and was confirmed at full budget.

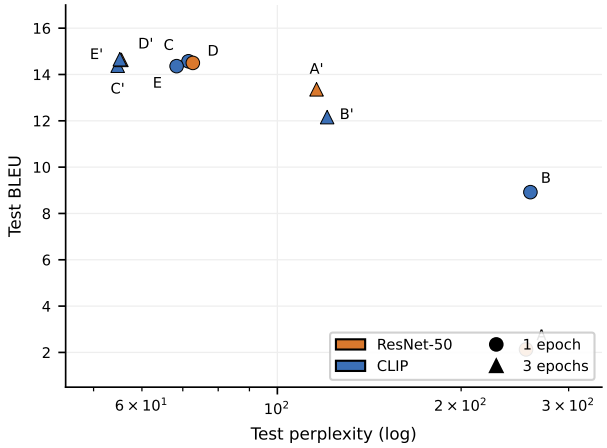


Figure 5. BLEU vs. perplexity (log x). Marker shape is the tier; colour is the encoder. Lower perplexity does not imply higher BLEU.

5. Discussion

Decoder choice is the critical decision. Across both budgets the Transformer decoder dominates. With an LSTM, CLIP helps substantially at the short budget, but convergence closes much of the gap at three epochs; with a Transformer, neither the encoder nor the embedding size materially changes BLEU. This is the central, checkable claim of the study.

Validation loss is a poor ranking signal. The screen–confirm protocol chose embed-768 by validation loss, yet at full budget the BLEU differences are negligible (14.36–14.66). Perplexity reflects teacher-forced fit, not generation quality; a BLEU (or human) signal is needed to choose models.

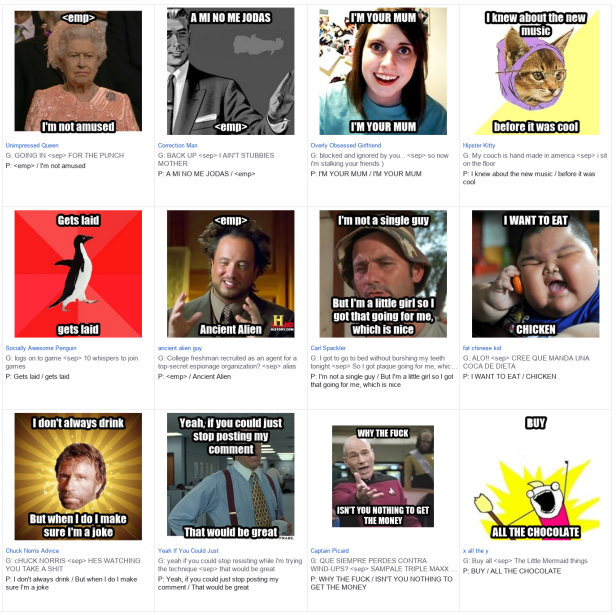
Encoder and decoder interact. At Tier 1 the decoder swap is large and the encoder swap is large only under the weak LSTM; at Tier 2 the encoder swap turns slightly negative and the decoder gap narrows, confirming that with enough training the encoder becomes nearly irrelevant behind a competent decoder.

6. Conclusion

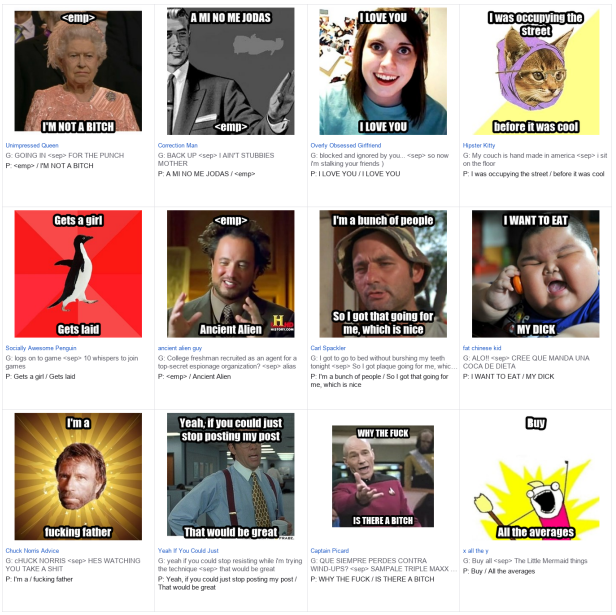
On memes900k, the decoder architecture dominates: Transformer decoders reach ∼14.5 BLEU versus 12–13 for converged LSTMs, and behind a Transformer decoder the encoder and embedding size wash out. Perplexity and BLEU decouple—halving perplexity leaves BLEU flat—the clearest diagnostic of exposure bias. The main contribution is methodological: test-set BLEU under a shared greedy decoder, a clean 2×2 isolation, and a two-tier budget that make the claims reproducible.

Future work. Scheduled sampling or REINFORCE to reduce exposure bias; an unseen-template split for true generalization; humor-aware metrics (CIDEr/ROUGE) while keeping BLEU for comparability; larger vision encoders or end-to-end fine-tuning; and systematic data scaling on the full 750,000-row training set.

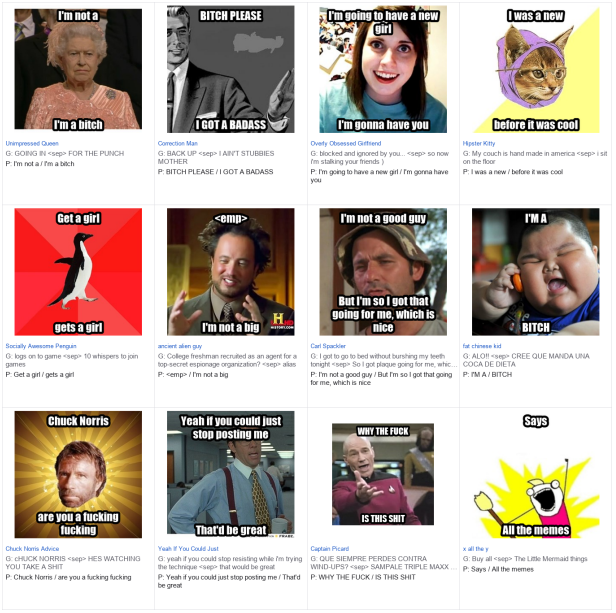
Qualitative examples. Figure 6 renders gold (G) versus generated (P) captions for four runs. Generated captions are fluent but often diverge from the reference, which is expected given the many valid captions per template and directly explains the low absolute BLEU.



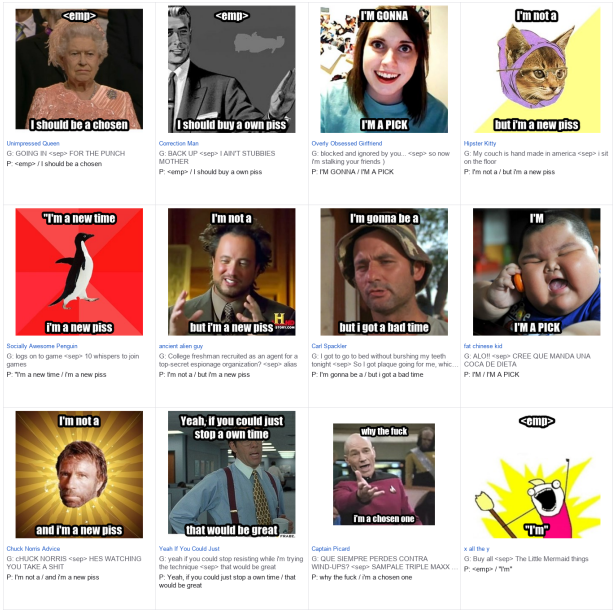
best run (CLIP+Transformer, 3 ep, emb 768)



ResNet-50+Transformer (3 ep)



ResNet-50+Transformer (1 ep)



CLIP+LSTM (3 ep)

Figure 6. Gold (G) vs. generated (P) captions for four runs. Generated captions are fluent but often diverge from the reference, which is expected given the many valid captions per template and directly explains the low absolute BLEU.

Limitations

- Seen templates only: the caption-level split shares templates across splits, so scores do not measure unseen-template generalization.
- BLEU is harsh for memes: many valid captions exist per template and jokes need not share surface forms; BLEU is meaningful mainly between runs.
- Perplexity is not comparable to DeepHumor: we use GPT-2 BPE (50k), whereas DeepHumor builds a dataset vocabulary.
- Exposure bias remains: teacher forcing and free-running generation decouple; we quantify but do not fix this.
- Frozen encoders and a small budget: we do not fine-tune the vision backbones and train on a 200,000-row subsample.

Ethical Statement

The memes900k dataset is used for non-commercial academic research under its intended use; no human annotators were involved and no new personal data were collected. All models are trained from scratch on frozen, publicly available pretrained features, so the compute footprint is modest (Apple Silicon MPS, tens of GPU-hours). We reuse the open-source DeepHumor codebase in line with its license and cite it accordingly. Meme generation can reproduce or amplify offensive stereotypes present in the source data; we therefore report aggregate metrics only and do not deploy the model.

AI Use Statement

Generative AI tools were used only for limited assistance with code and LaTeX formatting. The research design, experiments, analysis, and the writing of this manuscript were conceived and carried out by the human author, who remains fully responsible for the content, accuracy, and originality of this submission.

Software and Data

The dataset (memes900k) and the DeepHumor reference implementation are publicly available. Trained checkpoints, evaluation JSONs, and the figure/plot scripts are provided in the project repository; an anonymized link will be added for review.

Acknowledgements

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